

Concentration of a solution

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The concentration of a solution is the quantity of a solute that is contained in a particular quantity of solvent or solution.

Knowing the concentration of solutes is important in controlling the stoichiometry of reactants for solution reactions

Weight percent:

Weight percent means the weight in gram of the substance present in 100 gram of the sample. Such term is represented by w/W in 100 gram of the sample.

2%w/W NaCl means 2g NaCl is present.

Weight volume:

Weight volume means the weight in gram of the substance present in 100 mL of the solution. Such term is represented by w/V.

2% w/V NaCl solution means 2g NaCl is present in either 100mL of water or 100mL of the solution.

ppm:

This is an abbreviation for "parts per million" and it also can be expressed as milligrams per liter (mg/L). This measurement is the mass of a chemical or contaminate per unit volume of water.

ppb:

This is an abbreviation for "parts per billion" and it also can be expressed as milligrams per liter ($\mu\text{g/L}$). This measurement is the mass of a chemical or contaminate per unit volume of water.

Molarity (M)

Molarity is the the number of moles of solute present in exactly 1 L of solution.

Molarity = moles of solute / liters of solution.

The units of molarity are therefore moles per liter of solution (mol/L), abbreviated as M.

Example: What is the molarity of a solution made when water is added to 11 g CaCl_2 to make 100 mL of solution?

Solution:

$11 \text{ g CaCl}_2 / (110 \text{ g CaCl}_2 / \text{mol CaCl}_2) = 0.10 \text{ mol CaCl}_2$

$100 \text{ mL} \times 1 \text{ L} / 1000 \text{ mL} = 0.10 \text{ L}$

molarity = $0.10 \text{ mol} / 0.10 \text{ L}$ molarity = 1.0 M

Molality (m)

Molality is the number of moles of solute per kilogram of solvent.

Because the density of water at 25°C is about 1 kilogram per liter, molality is approximately equal to molarity for dilute aqueous solutions at this temperature. But, at very low or very high temperature this approximation is not true.

Example: What is the molality of a solution of 10 g NaOH in 500 g water?

Solution:

10 g NaOH / (40 g NaOH / 1 mol NaOH) = 0.25 mol NaOH

500 g water x 1 kg / 1000 g = 0.50 kg water

molality = 0.25 mol / 0.50 kg

= 0.50 m

Normality (N)

Normality is equal to the *gram equivalent weight* of a solute per liter of solution.

A gram equivalent weight or equivalent is a measure of the reactive capacity of a given molecule. Normality is the only concentration unit that is reaction dependent.

Example:

1 M sulfuric acid (H₂SO₄) is 2 N for acid-base reactions because each mole of sulfuric acid provides 2 moles of H⁺ ions.

Normality (N) = $Normality, N = \frac{\text{gram equivalent weight}}{1 \text{ liter of solution}}$ number of equivalents of solute
equivalents = 1 liter of solution liter

Gram-equivalent:

equivalent weight of an acid

The equivalent weight of an acid is the mass of acid required to generate one mole of hydrogen (hydronium) ion in aqueous solution. For an acid like HCl, which has only one ionizable hydrogen, the equivalent weight is the same as the molar mass.

Acid	Molar Mass	Equivalent Weight
HCl	36.5 g	36.5 g
H ₂ SO ₄	98.0 g	98.0g/2 = 49.0 g
H ₃ PO ₄	98.0 g	98.0g/3 = 32.7 g

The equivalent masses of acids, bases, and salts are calculated as follows: Equivalent mass of an acid = molecular mass of the acid/basicity. Equivalent mass of a base = molecular mass of the base/acidity. Equivalent mass of a salt = molecular mass of the salt/total positive valency of metal atoms.

Relation between normality and molarity:

We know normality = $\frac{\text{No. of gram-equivalent of the substance}}{\text{Volume in litre (V)}}$

$$= \frac{g}{E} \times \frac{1}{V} \left(\frac{g}{V} = \text{No. of gram - equivalent of the substance} \right)$$

$$\therefore \text{Normality} \times E \text{ (Equivalent weight)} = \frac{g}{V} = \frac{\text{gram}}{\text{litre}}$$

Similarly molarity = $\frac{\text{No. of gram-mole of the substance}}{\text{Volume in litre (V)}} = \frac{n}{V}$

$$\frac{g}{M} \times \frac{1}{V} \left(n = \frac{\text{weight of the substance in gram}}{\text{molecular weight}} \right)$$

$$\therefore \text{Molarity} \times M \text{ (Molecular weight)} = \frac{g}{V} = \frac{\text{gram}}{\text{litre}}$$

$$\text{Since Normality} \times E \text{ (Equivalent weight)} = \frac{g}{V} = \text{molarity} \times M$$

$$\therefore \text{Normality} \times E \text{ (Eq. wt.)} = \text{molarity} \times M \text{ (mol. wt.)}$$

$$\therefore \text{Normality} = \text{Molarity} \times \frac{M}{E}$$

$$\text{And Molarity} = \text{Normality} \times \frac{E}{M}$$